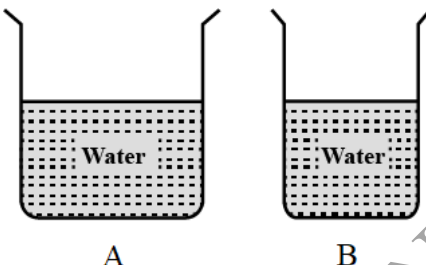
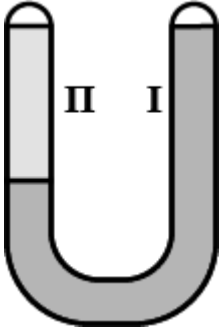


FREEDOM INTERNATIONAL SCHOOL
WORKSHEET
PHYSICS
MECHANICAL PROPERTIES OF FLUIDS

CLASS: XI

MAX MARKS: 10

TIME: 15 MINS.

1.	From the following figures, the correct observation is  A B (a) the pressure at the bottom of tank A is greater than that at the bottom of B (b) the pressure at the bottom of the tank A is smaller than that at the bottom of B (c) the pressure depends on the shape of the container (d) the pressure at the bottoms of A and B is the same.	1
2.	The radius of one arm of a hydraulic lift is four times of the radius of another arm. What force should be applied on the narrow arm to lift 100 kg? (a) 26.5 N (b) 62.5 N (c) 6.25 N (d) 8.3 N	1
3.	The ratio of the terminal velocities of two drops of radii R and R/2 is (a) 2 (b) 1 (c) 1/2 (d) 4	1
4.	A U-tube of uniform cross-section is partially filled with a liquid I. Another liquid II which does not mix with liquid I is poured into one side. It is found that the liquid levels of the two sides of the tube are the same, while the level of liquid I has risen by 2 cm. If the specific gravity of liquid I is 1.1, the specific gravity of liquid II must be  (a) 1.12 (b) 1.1 (c) 1.05 (d) 1.0	

5.	A mercury drop of radius 1 cm is broken into 10^6 droplets of equal size. The work done is ($\rho = 35 \times 10^{-2} \text{ N/m}$) (a) $4.35 \times 10^{-2} \text{ J}$ (c) $4.35 \times 10^{-6} \text{ J}$ (b) $4.35 \times 10^{-3} \text{ J}$ (d) $4.35 \times 10^{-8} \text{ J}$	1
6.	Dynamic lift is related to (a) Bernoulli's theorem (c) Equation of continuity (b) Archimedes' principle (d) Pascal's law	1
7.	The surface tension of a soap solution is $25 \times 10^{-3} \text{ N/m}$. The excess pressure inside a soap bubble of diameter 1 cm is (a) 10 Pa (b) 20 Pa (c) 5 Pa (d) none of these	1
8.	If the liquid neither rises nor falls in a capillary tube, then angle of contact is (a) 0° (b) 180° (c) 90° (d) 45°	1
9.	Water flows along a horizontal pipe whose cross-section is not constant. The pressure is 1 cm of Hg, where the velocity is 35 cm/s. At a point where the velocity is 65 cm/s, the pressure will be (a) 0.89 cm of Hg (c) 0.5 cm of Hg (b) 8.9 cm of Hg (d) 1 cm of Hg	1
10.	For different capillaries of radius r, the condition of the rise in liquid (h) above the liquid surface is (a) $rh = \text{constant}$ (c) $h + r = \text{constant}$ (b) $h/r = \text{constant}$ (d) $h - r = \text{constant}$	1